E26: equivalent encodings and matched ordinary controls

2026-10-25; bounded public toy fixtures; Intel Xeon Platinum 8573C; managed Linux x86_64; isolation L0. No admitted wall time.

The native construction covers all 90 preregistered workloads: ten ordinary curves, B=8,12,16 abscissae, arities 2,3,4 and original target label 0. All four candidates and the ordinary controls are retained. The requested solver population is 631 cells including one A/A control. This is a partial E26 pilot, not complete IC costing or a speedup claim.

Verified construction and information checks

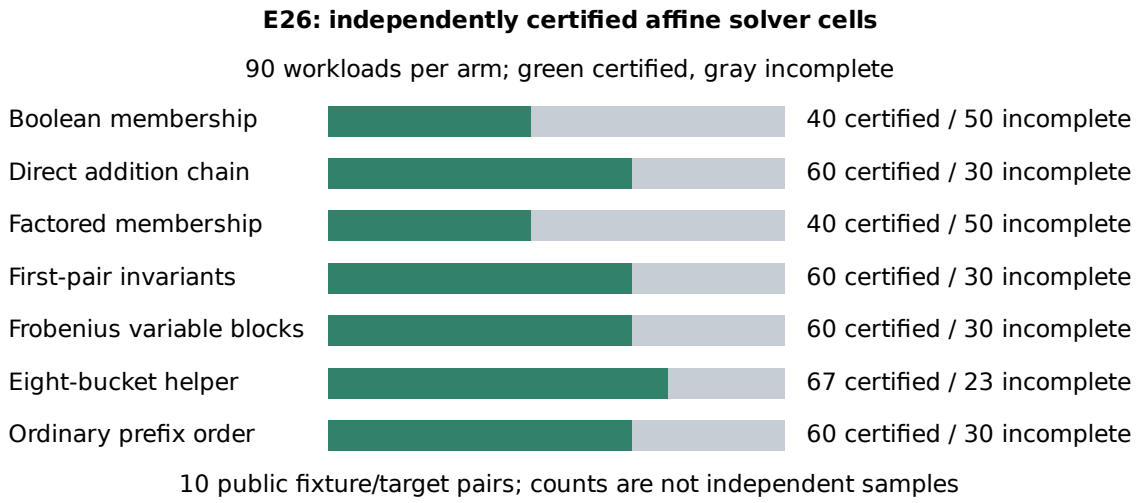
Independent PARI replay accepts 18970 retained signed-witness checks and rejects 18623 wrong-sign controls. The Rust gate checks the script and output hashes, completion marker and clean errors. Counts repeat point tuples across overlapping base sizes. There are 10 distinct fixture/target pairs.

All 90 matched-order checks recover the ordinary polynomial coefficients exactly after setting anti-rational variables to zero. This is an input-level identity check for the frozen family, not a universal statement about every twist encoding. The invariant diagonal and identity-prefix exceptions remain in the construction and full relation census.

The same-input A/A polynomial check passes; equal independently certified row-multiplication counts: **true**. Native step-log counter coverage failures: **0**. Any failed or missing certificate remains incomplete.

Complete and incomplete solver cells

Arm	Requested	Certified	Incomplete	msolve branches certified
Boolean membership	90	40	50	40
A/A control	1	1	0	1
Direct addition chain	90	60	30	60
Factored membership	90	40	50	40
First-pair invariants	90	60	30	120
Frobenius variable blocks	90	60	30	60
Eight-bucket helper	90	67	23	536
Ordinary prefix order	90	60	30	60



Native F4 uses grevlex, degree bound 32, GM=1 and a shared 30-second budget per candidate cell, including every helper/diagonal branch. Stock msolve q0 and Singular share a separate 30-second cell budget. The native and external phases each have a 1,800-second global cap, with checkpoints. Global-cap omissions remain not attempted. Every process has one worker and a 4 GiB address-space cap. Raw failure outputs and exact inputs are retained.

Singular computes a reference basis and checks both ideal containments, equality of leading ideals and vdim against the complete native affine assignment census. These are complete-basis checks. vdim is the quotient vector-space dimension, not Krull dimension. Parsed stock msolve bases have their own independent certificate; an msolve failure does not erase a valid native certificate. The small positive/truncated-basis control is retained.

Fixed first workload: what the recorded counts measure

The p=59, B=8, arity-2 example was fixed before inspecting outcomes. Counts below are multiplications in native row reduction only. A smaller number is not a measured full-solver or IC speedup. Incomplete counter sums are partial records, not zero-cost successes.

Encoding	Recorded row multiplications	Basis status
Direct addition chain	7920	certified
Ordinary prefix order	7920	certified
Frobenius variable blocks	7920	certified
Factored membership	473345	certified
Boolean membership	447400	certified
Eight-bucket helper	458	certified
First-pair invariants	2594	certified

Construction, symbolic preprocessing, normalization, root extraction, process startup, signs, deduplication, exception enumeration, transfer, verification and complete target recovery are not priced by this counter. All eight helper solves are included in its row-multiplication sum, while their remaining overhead is missing. No admitted wall time is reported. The ordinary comparison is the preregistered hindsight minimum among four fixed controls; it is not an executable selector or an IC pipeline baseline.

Whole-worker instruction diagnostics

Callgrind records simulated guest instructions Ir, not cycles, wall time, field operations or GAE. The first probe starts a separate process for each helper/diagonal branch. All 31 native and stock-msolve worker outputs match the previously certified bases (and native row counters). The second probe runs every native candidate through the same batch interface, one process per candidate; all eight process outputs and every branch match. Independent audits reopen the raw Ir records and recompute their aggregates.

Encoding	Native separate Ir	Native batch Ir	msolve separate Ir
Boolean membership	21850152	21911730	25983653
Direct addition chain	2423541	2431742	11030733
Factored membership	63425225	63477695	34379495
First-pair invariants	2324142	1847549	20283359
Frobenius variable blocks	2416229	2436407	10980407
Eight-bucket helper	5784840	2458250	78063100
Ordinary prefix order	2422113	2431752	10976860

Fixed p=59 case: simulated guest instructions Ir

Encoding	Native separate	Native batch	msolve separate
Boolean membership	21,850,152	21,911,730	25,983,653
Direct addition chain	2,423,541	2,431,742	11,030,733
Factored membership	63,425,225	63,477,695	34,379,495
First-pair invariants	2,324,142	1,847,549	20,283,359
Frobenius variable blocks	2,416,229	2,436,407	10,980,407
Eight-bucket helper	5,784,840	2,458,250	78,063,100
Ordinary prefix order	2,422,113	2,431,752	10,976,860

Different process models; no admitted time or complete IC cost

Separate-worker A/A relative spread is 0.0015427351256755144; batch-worker A/A relative spread is 0.0000625085846493720 8. Both meet the frozen 5-percent check. These are same-input repeats, not confidence intervals or an independent-host replication.

The helper's 458 row multiplications omit enough other work to change its ranking: separate helper workers use 5,784,840 Ir versus the direct worker's 2,423,541. Batching the helper branches records 2,458,250 Ir versus the matched direct batch's 2,431,742. The pair-invariant batch records 1,847,549 Ir on this fixed case; its stock-msolve separate-worker total is 20,283,359 Ir versus direct's 11,030,733. Process model and solver therefore matter. These are observations of this prototype, not an inherent ranking or a twist-specific advantage. No fastest-implementation claim is made.

The worker interval includes startup, input parsing, symbolic preprocessing, F4 and output serialization. It excludes encoder construction, parent orchestration, OS/kernel work, certification, root/sign lifting, duplicate removal, linear algebra over relations and target recovery. The batch follow-up was motivated by the first probe's training result; it is not heldout. There is no batched msolve measurement. Complete-cost and calibrated IC comparisons remain open.

Tuple multiplicity, coefficient rows and rank

Frozen workload	Tuple witnesses	Distinct rows	Rank mod source order
p=59, B=8, m=2	1	1	1
p=211, B=8, m=3	17	6	5
p=251, B=16, m=4	1847	162	16

The native auditor deduplicates signed coefficient vectors and computes rank over the prime source subgroup order, rather than the curve coordinate field. A separate PARI input replays all 90 ranks. Full-task ranks include exceptional relations; they are not used as denominators for affine-solver cost. Exactly one retained deterministic sign witness per canonical tuple is used, not every possible sign assignment. The first-pair permutation is folded consistently.

Scope, sources and remaining work

This wave implements and runs bounded E26 encodings. The genus-two transfer result remains the prior-art BHLS construction with 24 verified toy targets; no new transfer-speed result is claimed. Earlier extension/rational helper row equalities remain valid. The 270-cell E30 pilot is a separate experiment, and its full solver-cost comparison is still open.

E26 remains partial: broader training/heldout cases, production root extraction, matrix nonzeros, complete counters and full costs are missing. E27 complete cold accounting/calibration and E28 final selection are unrun; these worker probes do not replace either requirement. Labels 32--63 remain unopened. No novelty, fastest-tool, NIST DLP, independent-host, calibrated IC speedup or leaderboard promotion follows. All cards remain blocked and unfunded; no paid host or lease was created.

Primary assessment, independent audit, batch instruction audit, separate-worker probe, batch-worker probe, protocol, matched-control amendment, audit protocol, worker protocol, batch protocol, raw artifact manifest, validation logs.

The new raw bundle contains 12707 archived files and 39 separately tracked metadata records. Its 9310540-byte immutable object is published in the user-confirmed Setec bucket. Fresh download, remote verification and every member hash pass; extraction/repacking reproduces the exact archive digest. Publication receipt and downloaded-member check. The native index entry accompanies these records. The separate delivery receipt tracks the pushed head and CI without changing the scientific measurements. The earlier archive and index entry are unchanged.

Exact source, native/tool binary hashes, fixture and input hashes, commands, caps and raw outputs are retained with the records. UTC intervals are retained where captured. The batch probe's launch timestamps were not captured; its post-run receipt records that gap and the saved actual child exit explicitly. GitHub access initially returned 403/404. A later CLI fetch and PR read succeed and resolve the existing PR to isogeny-labs/crypto; the connector's old path still returns 404. Delivery and its exact head are tracked separately in the access problem. The twelve historical oversized Git blobs remain a separate merge hold. The IC leaderboard, progress timeline and scientific geometry figures remain unchanged because this wave establishes no complete-cost result. Benchmark copies are indexed as unadmitted diagnostics without adding a validated row.

Requirement status

Requirement	Status	Remaining acceptance gap
Genus-two route	Partial; earlier 24 toy transfers unchanged	Full transfer and destination-solving costs; polarized graph identification
Extension helpers	Tested families retain matched rows	A new restriction or cheaper solver computing the same projected task
E26 encodings	Bounded pilot verified; full experiment partial	Broader training/heldout, root extraction, nonzeros and complete costs
E30 finite incidence	270 correctness cells verified earlier	Separate full solver-cost comparison remains unrun
E27 complete cold cost	Unrun; worker probes are separate diagnostics	Original 24-arm instrumentation and all 1,920 attempts, calibrated matched rho
E28 heldout selector	Unrun; labels 32–63 unopened	Freeze selector only after complete training and E27 counters
Delivery	Tracked by separate commit/push receipt	Current-head CI/review; historical size hold remains

Next discriminating experiments retain the original blocked cards and distinguish proposed runs from this completed bounded wave. No subsequent job is implied to be running.